Spectral Spatial Window Attention Transformer for Hyperspectral Image Classification

Xi Wu, Tahir Arshad, and Bo Peng

***Abstract*— Hyperspectral image (HSI) data have a wide range of spectral information that is valuable for numerous tasks. HSI data encounter some challenges, like insufficient representation of spectral spatial data, data scarcity, and redundant information. Researchers introduce various research works to address these challenges, with convolutional neural networks (CNNs) finding widespread use in HSI classification because of their capacity to extract features from HSI data. Due to a limited receptive field, CNN cannot handle long-range dependencies. To overcome the above-mentioned issue, we proposed a cross-window spectral spatial transformer model for HSI classification. First, the fused-MBConv module is used to extract low-level spectral spatial features from HSI data. Second, we propose cross cross-window attention module to model the long-range spatial dependencies and obtain refined spectral spatial features. Finally, we proposed an interactive feature-enhanced (IFE) module to obtain global contextual features, leading to more discriminative representations. We tested the proposed model on four publicly available HSI datasets: Xuzhou, Salinas Pavia University, and WHU-Hi-HC. Using only 0.5% and 0.05% of the training samples from the four datasets, we achieved the best classification in terms of overall accuracy (OA), average accuracy (AA), and Kappa coefficient.**

***Index Terms*— Attention module, convolutional neural network, (CNN), hyperspectral image (HSI) classification, super resolution, vision transformer (VIT).**

I. Introduction

THE hyperspectral image (HSI) data have numerous narrow bands containing a substantial amount of information. HSI classification is a part of Earth observation [1] extensively used in agriculture [2], mineralogy [3], and environmental sciences. HSI classification data captures spectral information from numerous contiguous spectral bands of surface objects. In the HSI classification, there are three main challenges. First, in the spectral dimension of HSI data, hundreds of bands exist and have a lot of information. Still, the bands' values and information are frequently redundant, which requires a lot of computational costs. Second, in HSIs, pixels are mixed, interfering with the classification since one pixel usually corresponds to a different object category. Third, manually labeling HSI samples is costly and time-consuming, leading to limited availability of labeled samples. Numerous approaches have been proposed in the last decade to address these challenges. This work proposes to solve the problem of limited training samples in HSI data. Initially, researchers devised traditional methods of machine learning for the categorization of HSIs, such as k-nearest neighbor [4], logistic regression [5], Bayesian estimation [6], and support vector machines [7]. Still, it has been noted that these traditional categorization models frequently lead to misclassification. Furthermore, other techniques have been devised to reduce dimensionality and retrieve spectrum information, including principal component analysis (PCA) [8] and linear discriminative analysis (LDA) [9].

However, these techniques often fail to consider the spatial correlation between pixels in a spatial dimension, which is essential for achieving effective spatial feature extraction. To tackle this problem, several mathematical operators have been devised, including the morphological profile [10], [11]. Deep learning is well well-recognized and highly successful technique for extracting features in the context of hyperspectral classification [12]. For instance, a stacked autoencoder (SAE) [13], a deep belief network (DBN) [14] involves the extraction of depth and invariant characteristics from hyperspectral data. The obtained characteristics are then utilized in logistic regression to tackle the challenge of HSI categorization. However, it is crucial to acknowledge that both (SAE) and (DBN) encode spatial information as vectors during the pretraining phase, leading to the unavoidable loss of spatial information.

It is important to mention that a convolutional neural network (CNN) [15], [16]. References [17] and [18] introduced a HybridSN network to simplify the net framework by integrating spectral–spatial information using 3-D and 2-D convolutions. The basic hybrid model outperforms both the 3D-CNN model and the 2D-CNN model in terms of computing efficiency, and it demonstrates superior performance in handling the issue of small sample size. A spectral–spatial capsule network proposed by Paoletti et al. [19] to reduce the complexity of convolutional networks. Some recent works [20], [21], [22] investigate distinctive methods to enhance the discrimination of spectral–spatial features. The attribute profiles of Aptoula et al. [23] are utilized to provide the spectral and geometric features of HSI for the CNN. To address the issue of inadequately labeled samples, Li et al. [20] proposed a data augmentation technique. Chen et al. [21] proposed to combine

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a CNN with Gabor filters to enhance texture and edge information. Mei et al. [24] utilized a CNN to identify sensor-specific features for HSI. Li et al. [25] improved spectral–spatial features by utilizing PCA and a fully convolutional network. The attention mechanism is inspired by the human visual system to identify important regions from images for classification [26]. The attention recurrent convolutional network becomes excessively complex when combining LSTM and CNN to extract attention features. A two-branch attention module proposed by Haut et al. [26], where one branch is utilized to generate an attention mask and the other branch is utilized to extract convolutional features. Then, by multiplying the attention mask with the convolutional attention features, are obtained. Different from earlier methods, an attention module and an interactive feature enhancement module (IFE) that integrated into a simple component. To extract attention features, the suggested feature enhancement module makes use of the correlation between the hyperspectral pixels inside an HSI cube.

Recently vision transformers (ViTs) [27] architecture is based on self-attention mechanisms combined with multilayer perceptrons (MLPs), which excel at capturing long-range dependencies between tokens in a sequence. Originally designed for machine translation tasks in natural language processing [28], ViTs have gained significant attention due to their remarkable success, prompting researchers to explore their application in image processing and computer vision. In contrast to convolutional networks, transformers offer a fresh perspective and new possibilities for improving performance at the attention level. The key factor behind the transformer's success lies in its ability to effectively model long-range relationships, overcoming limitations faced by traditional architectures. The transformer has been recently introduced in HSI classification. Hong et al. [29], He et al. [30] treated HSI patches as sequential data from the spectral dimension, utilizing group-spectral embedding and a transformer encoder module to capture locally detailed spectral representations. By integrating both global and local information, this approach significantly enhances the model's perceptual capabilities. Sun et al. [31] introduced the spectral–spatial feature tokenization transformer (SSFTT) [31], designed to capture both spectral–spatial and high-level semantic features. The model initially extracts shallow spectral and spatial features using a combination of 3-D and 2-D convolutional layers. It then applies a Gaussian-weighted feature tokenizer to transform these features, which are subsequently fed into a transformer encoder for enhanced representation and learning. Ma et al. [32] proposed a ViT incorporating a lightweight self-Gaussian attention (LSGA) mechanism to extract global deep semantic features. Yang et al. [33] introduced an HSI transformer that incorporates two key modules: spectral adaptive 3-D convolution projection and convolution permutator, designed to capture subtle spectral and spatial information. Li et al. [34] proposed a multigranularity ViT using a semantic tokens transformer to learn multigranularity features and enhance accuracy. They employed the LFE module to extract local features. Ouyang et al. [35] developed the HybridFormer network, which combines CNNs for extracting shallow features with a spectral–spatial attention (SSA) based transformer encoder for learning semantic features. Transformers designed for HSI classification represent spatial patches or spectral bands as tokens. While these tokens enable the modeling of intricate relationships across spatial and spectral dimensions, the high token count results in substantial computational overhead.

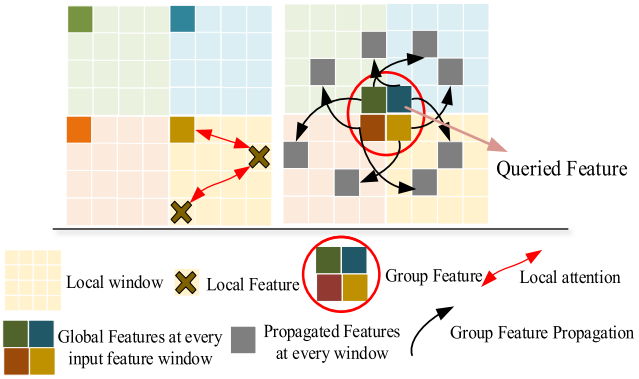


Fig. 1. Illustration of propagation of the group features. The group features is propagated between the local windows to model long-range dependencies.

In this study, while many models successfully extract spectral–spatial information from HSIs, despite the strength of transformer-based models, they often come with high-computational demands and may struggle to efficiently capture fine-grained spatial details and scale variations, resulting in suboptimal performance in HSI classification. Fig. 1 represents the idea of this article, where we proposed cross-window attention to learn the global dependencies. First, the feature grouping is applied, and then the grouped features are propagated between the windows to model long-range spatial dependencies. The main contribution of this article is as follows.

1) The proposed framework is designed for HSI classification. It utilizes a combination of a spectral spatial feature extractor with cross-window attention and an interactive feature-enhanced module. This model presents a novel approach for precise image classification in hyperspectral remote sensing, exhibiting its effective extraction capabilities for spatial–spectral and contextual information in HSIs.
2) Cross-window attention mechanism modeling the long-range spatial dependencies to group the features and propagating to the feature windows. That enables the model to focus on the most informative pixels to enhance the accuracy and efficacy of the classification process.
3) An efficient, IFE is proposed to extract global contextual features and also maintains the interaction between high-level semantic features to improve the classification accuracy.

The rest of this article is organized as follows: Section II explains related work; Section III describes the Proposed Model, Section IV outlines the Dataset and Experiment Evaluation; Section V Results and discussion, and Section VI conclusion.

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II. Related Work

A. CNN-Based Methods for HSI Classification

Researchers have utilized CNNs for HSI classification due to their ability to automatically extract features, eliminating the need for complex image preprocessing. Compared to traditional methods, CNN-based approaches deliver superior classification performance. Based on the feature dimensions, they process, CNN-based methods can be classified into three categories: 1D-CNN, 2D-CNN, and 3D-CNN.

In the early stages of HSI classification, 1D-CNNs were primarily utilized to extract feature vectors along the spectral dimension. For instance, Hu et al. [15] proposed an HSI classification model employing a 1D-CNN with five convolutional layers. Compared to 1D-CNNs, 2D-CNNs offer the advantage of capturing spatial context, effectively modeling spatial dependencies within HSI data. In [11] and [36], 2D-CNN models were developed to extract spatial features, following a dimensionality reduction step using PCA. Additionally, in [37], a dual-branch network was introduced, integrating both 2D-CNN and 1D-CNN architectures, where the 2D-CNN focused on spatial feature extraction while the 1D-CNN specialized in spectral feature extraction for enhanced HSI classification performance.

To better utilize the intrinsic characteristics of HSI data, joint extraction of spatial and spectral information, rather than separate processing, is considered a more effective approach. Consequently, 3D-CNNs have been widely adopted for HSI classification. Chen et al. [38] introduced a 3D-CNN framework incorporating virtual sample enhancement and regularization techniques, achieving promising classification results. Furthermore, inspired by the pyramidal hierarchical structures commonly used in CNNs, Rao et al. [39] proposed the 3-D Adaptive spatial–spectral pyramidal layer CNN model (ASSP-SCNN). This model leveraged multiscale samples during training to fully exploit spatial–spectral features while mitigating overfitting, thereby enhancing classification performance. Xu et al. [40] were the first to propose feeding the entire image directly into the model, rather than inputting pixel patches one at a time, to learn global representations. Building on this idea, Wang et al. [41] introduced FullyContNets, which also utilize the full image and adaptively aggregate multiple features through a pyramid multiscale structure. Experimental results demonstrated that leveraging the rich spatial information from the complete image can lead to excellent HSI classification performance.

B. Transformer-Based Methods for HSI Classification

Recently, transformers have demonstrated remarkable potential across a wide range of tasks. Unlike CNNs, transformers employ a fundamentally different architecture based on the attention mechanism and MLPs. This design enables transformers to effectively capture global dependencies, making them highly suitable for complex pattern recognition tasks. The introduction of the ViT [27] has further facilitated the adoption of transformers in various computer vision applications, establishing them as a powerful alternative to traditional deep learning models. The Swin Transformer [42] is a ViT architecture proposed for computer vision tasks, image classification, and segmentation. This design allows for improved efficiency and scalability in handling high-resolution data. Swin Transformer restricts self-attention to nonoverlapping local windows, greatly reducing complexity to linear with image size. To enable cross-window connections and capture long-range dependencies, it introduces a shifted window mechanism, where the window partitions are shifted between layers. Swin Transformer builds a hierarchical representation by merging patches at different stages, similar to the multiscale features in CNNs. Swin Transformer achieves efficient and scalable feature extraction by combining local self-attention, cross-window communication, and a hierarchical structure. Among the pioneering studies applying transformers to HSI classification, Hong et al. [29] treated spectral signatures as sequential data and tokenized preprocessed local spectral features into spectral tokens. These tokens were then processed by a transformer encoder with cross-layer skip connections, facilitating effective feature learning. Recently, Roy et al. [43] proposed MorphFormer, a novel model that integrates spectral and spatial morphological convolution with transformers. This approach enhances the interaction between image tokens and class tokens, effectively capturing structural and shape information in HSI data. For instance, [44] applies FFT repeatedly within nonhierarchical frameworks, resulting in redundant computational complexity without efficiently transitioning from shallow to deep feature representations. Zhao et al. [45] introduced the groupwise separable convolutional ViT (GSC-ViT), designed to efficiently capture both local and global features while providing a more lightweight and computationally efficient solution. Mei et al. [46] introduced a group-aware hierarchical transformer network (GAHT) incorporating a grouped pixel embedding module, which enhances the model's capability to capture local relationships within HSI spectral bands while preserving a global perspective of the spatial–spectral context. Qiao et al. [47] introduced an innovative network termed the multiscale neighborhood attention transformer (MSNAT). This approach focuses on neighborhood pixels within local windows and employs varying window sizes to capture multiscale spatial information, leading to a notable enhancement in HSI classification performance. CNNs excel in processing local features, while ViTs offer a global receptive field, each providing unique strengths to their respective models. Combining CNN and ViT results in a model with powerful feature extraction capabilities and recognition performance, as demonstrated by earlier studies. Leveraging this integrated CNN-ViT architecture, we proposed a spectral spatial window attention transformer for HSI classification.

III. Proposed Methodology

Fig. 2 depicts the proposed model. This section will elucidate the model's structure and its functioning. In this study, we try to solve the problem of a small training sample through network structure. First, a dimensionality reduction technique using PCA is applied to the raw HSI data. Let the HSI data be represented by $X_{\text{hsi}} \in \mathbb{R}^{H\times W\times C}$, where $H \times W$ indicates the height and width of the data, and C

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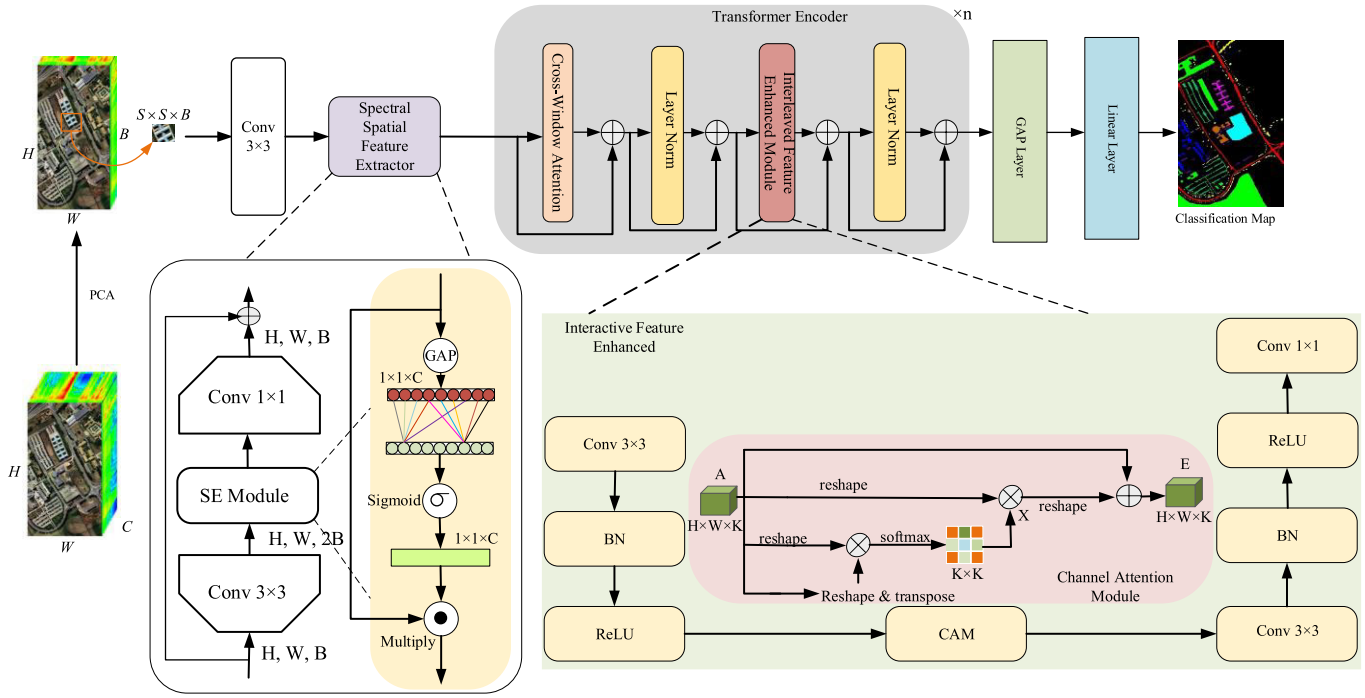


Fig. 2. Illustration of the proposed model for HSI classification.

represents the spectral dimension. After applying PCA on the spectral dimension, the hyperspectral data can be represented as $X_{\text{hsi}} \in \mathbb{R}^{H\times W\times B}$. Suppose that X_{hsi} contains N labeled pixel $X = \{x_1, x_2, \ldots, x_N\} \in \mathbb{R}^{1\times 1\times C}$ and their corresponding one hot label $Y = \{y_1, y_2, \ldots, y_N\} \in \mathbb{R}^{1\times 1\times K}$ where K represents the number of classes. The spatial size $S \times S$ around center pixel can be defined as a spectral spatial vector. The difference between the predicted class label and the ground-truth label is estimated through cross-entropy loss function.

A. Pectral Spatial Feature Extractor

Convolution procedures are the most significant computational cost in CNNs. Let $X^l \in \mathbb{R}^{H\times W\times C} l$ is 2-D convolution layer $H \times W$ represent the height and width of the feature maps, B represent the channel. Given the 2-D convolution kernels ($k_h \times k_w$), and the output channel is C_{out}, the output feature maps is $X^{l+1} \in \mathbb{R}^{H_{\text{out}}\times W_{\text{out}}\times C_{\text{out}}}$. For the spectral spatial feature extraction module, we used the fused-MBConv [48], [49] block. This model consists of a 3 × 3 depth convolution expansion layer, a squeeze and excitation module [50], and the 1 × 1 projection layer. The expansion ratio of the convolution is set to 2, and the 1 × 1 projection layer reduces the channel dimension by the same ratio. The fused-MBConv module is efficient in encoding the spatial features. Fig. 2 shows the fused-MBConv spectral spatial feature extractor module.

B. Cross-Window Attention

Fig. 3 shows the cross window attention module [51]. Cross-window attention refers to the exchange of information between neighboring windows, enabling the capture of long-range dependencies beyond local window boundaries. Unlike standard window-based attention like MSNAT, where interactions are confined within fixed windows, our method explicitly facilitates feature interactions across multiple windows. This cross-window behavior is reflected through feature grouping, where tokens are collected from different windows and concatenated; the use of an MLP-Mixer module for global information exchange; and cross-window propagation that allows information to flow between spatial groups, connecting non-adjacent regions. Thus, through grouping, ungrouping, and intermediate mixing operations, our model achieves cross-window attention rather than merely local window-based attention.

1) Feature Grouping: After the feature extractor module, the input of the group attention consists of image features $X \in \mathbb{R}^{N\times C}$ where N is the total number of image features and C represents the dimensionality of the feature vector. The matrix $G \in \mathbb{R}^{M\times C} S$ is utilized to store M learnable group tokens. The process of grouping is executed by a simple multihead attention process. This operation results in a generation of grouped features denoted $Y \in \mathbb{R}^{M\times C}$

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{1}$$

$$Y = \text{concat}_{\{h\}}\left(\text{Attention}\left(W_h^Q G_h, W_h^K X_h, W_h^V X_h\right)\right). \tag{2}$$

In the attention operation, the channel number is denoted as d, the head index is represented by h, and the projection matrices for the Q, K, and V are denoted as $W_h^{\{Q,K,V\}}$. after the concatenation procedure, we eliminate the feature projection layers and designate them as an identity matrix. Therefore, the grouped features can be described as the weighted summation of image features at each head, where the weights are computed using the attention operation.

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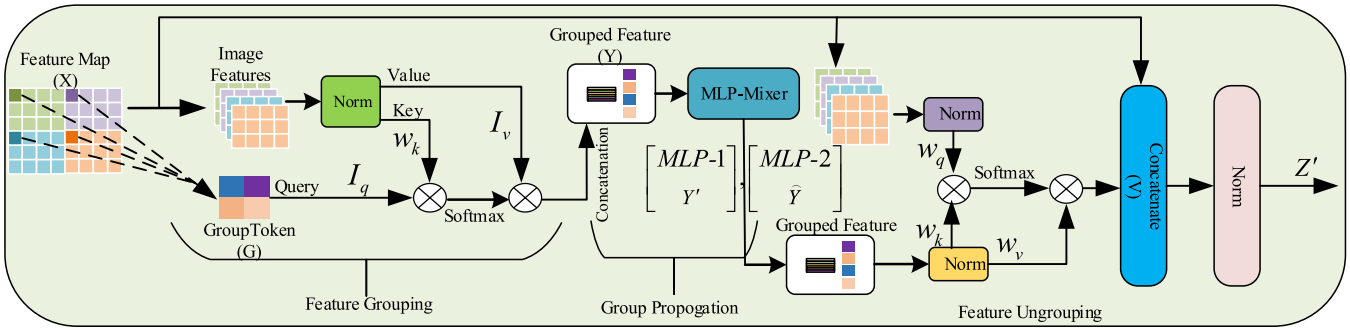


Fig. 3. Illustration of the cross-window attention module.

2) Group Feature Propagation: Once the grouped features have been obtained, it is possible to update and pass on global information among them. MLP-Mixer requires an established input size that aligns with our selected number of groups. Our MLP-Mixer comprises a set of sequential MLP models. $Y \in \mathbb{R}^{M\times C}$ contains the grouped features from the feature grouping process, which can be updated to $\hat{Y} \in \mathbb{R}^{M\times C}$ with the MLP-Mixer layer

$$Y' = Y + \mathrm{MLP}_1(\mathrm{LayerNorm}(Y)^T)T \tag{3}$$

$$\hat{Y} = Y' + \mathrm{MLP}_2(\mathrm{LayerNorm}(Y')). \tag{4}$$

The first MLP is employed for inter-group information mixing, whereas the second MLP is utilized for channel maxing. Where T is the transpose operation between token and channel mixing in the MLP Mixer layer.

3) Feature Ungrouping: Once the grouped features have been updated, it is possible to retrieve global information from the image features by employing a Feature Ungrouping procedure. The features are ungrouped by the utilization of a transformer decoder layer, wherein the image features query the grouped features

$$V = \mathrm{Concat}_{\{h\}}\left(\mathrm{Attention}\left(\tilde{W}_h^Q X_h, \tilde{W}_h^K \hat{Y}_h, \tilde{W}_h^V \hat{Y}_h\right)\right) \tag{5}$$

$$Z' = W_{\mathrm{proj}} * \mathrm{Concat}(V, X),\ Z'' = Z' + \mathrm{MLP}(Z') \tag{6}$$

where $\tilde{W}_h^{\{Q,K,V\}}$ is the projection matrix of the attention where W_{proj} represents the linear matrix used to project the concatenated features Z' onto the same dimension as the image features X. The original transformer decoder layer is altered by replacing the initial residual connection with a concatenation operation. Afterward, the feature projection layer is moved to transform the feature back to its initial dimensions.

C. Interactive Feature-Enhanced Module

To highlight the complex feature representation of the HSI data, we incorporate a convolution block into the IFE module, as illustrated in Fig. 2. This block effectively learns the interdependence among the high-level feature maps. To optimize feature interaction and efficiently understand nonlinear relationships during the feature integration stage. In practical terms, the IFE module first adopts a convolutional layer, and then, to speed up convergence, a BN layer is applied. Then, to improve the representation of the nonlinear feature, we employ the rectified linear unit (ReLU) activation function. Most significantly, feature maps are weighted, and interested regions are highlighted using a channel attention module (CAM). The expression for the IFE module is

$$\mathrm{IFE} = \mathrm{CBR}\big(\mathrm{CAM}\big(\mathrm{CBR}\big(Z'\big)\big)\big) \tag{7}$$

$$\mathrm{CBR}(\cdot) = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\cdot))) \tag{8}$$

where Z' is the output of the previous module, and CBR consists of a convolution layer, followed by a batch normalization layer, and a ReLU activation function layer. Let's assume the input feature maps of the CAM is $X_a \in \mathbb{R}^{H\times W\times K}$ first reshaped the feature into X_a to $X_b \in \mathbb{R}^{K\times(H\times W)}$ and $X_c \in \mathbb{R}^{(H\times W)\times K}$

$$X_{ji} = \frac{\exp\big(X_{bi}\cdot X_{cj}\big)}{\sum_{i=1}^{K}\exp\big(X_{bi}\cdot X_{cj}\big)}. \tag{9}$$

Reshape the X_a to $X_d \in \mathbb{R}^{K\times(H\times W)}$. The final result can be expressed as A softmax layer is applied to obtain the interact feature map: where X_{ji} represents the degree of correlation between the ith and jth channel feature

$$X_j = \partial\sum_{i=1}^{K}(X_{ji}X_{di}) + X_{aj} \tag{10}$$

where ∂ a learnable parameter, which is initialized to 0.

IV. Dataset and Experimental Evaluation

To verify the performance of the proposed methodology, three HSI classical datasets were selected for experiments, including the Xuzhou dataset, WHU-Hi-HanChuan, WHU-Hi-HongHu, and Salinas dataset. Characteristics of the datasets and information are summarized in Table I. All the datasets were preprocessed following the convention to remove noisy and water absorption bands. The application of PU dataset in agriculture, urban planning, environment monitoring, and material identification due to different classes. The application of the Salinas dataset in agriculture monitoring. The mineral classification is conducted on the Xuzhou dataset. The number of Training and testing samples for the experiments is further shown in Table II.

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TABLE I
DATASET NAMES AND RELATED INFORMATION

Dataset	Image size (Pixel)	Band	class	Spatial Resolution	Spectral Resolution
XU	500×260	436	9	0.73m	415-2508nm
PU	610×340	103	9	1.3m	430-860μm
SA	512×217	224	16	3.7m	400-2500μm
WHU-Hi-HC	1217×303	274	16	0.109m	400-1000nm

TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES ACROSS ALL DATASET

Class	Training Samples				Testing Samples			
	XU	PU	SA	WHU-Hi-HC	XU	PU	SA	WHU-Hi-HC
C1	131	33	10	45	26265	6598	1999	44690
C2	20	93	18	23	4007	18556	3708	22730
C3	14	10	9	10	2769	2089	1967	10277
C4	26	15	6	6	5188	3049	1388	5347
C5	66	6	13	2	13118	1339	2665	1198
C6	13	25	19	5	2523	5004	3940	4528
C7	35	6	17	6	6955	1324	3562	5897
C8	24	18	56	18	4753	3664	11215	17960
C9	15	4	31	9	3055	943	6172	9460
C10	–	–	16	11	–	–	3262	10505
C11	–	–	5	17	–	–	1063	16894
C12	–	–	9	4	–	–	1918	3675
C13	–	–	4	9	–	–	912	9107
C14	–	–	5	18	–	–	1065	18542
C15	–	–	36	2	–	–	7232	1134
C16	–	–	9	75	–	–	1798	75326

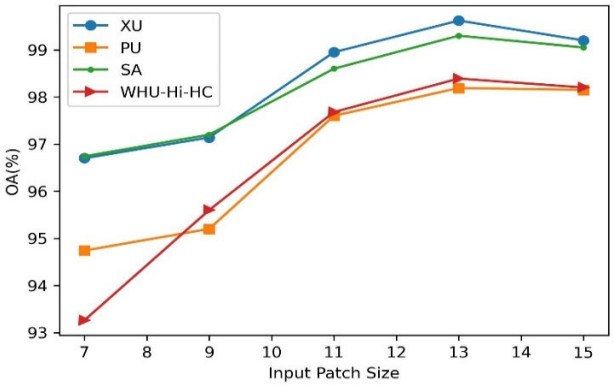


Fig. 4. OAs for three datasets with different patch sizes.

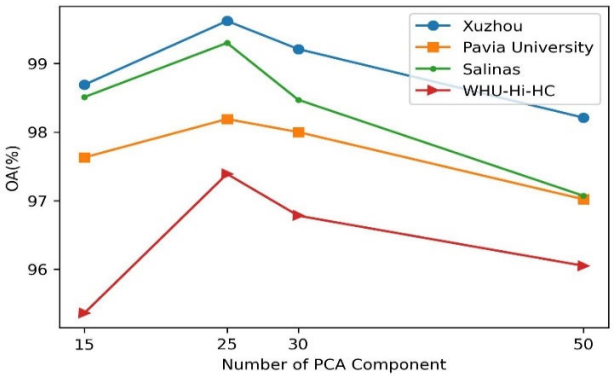


Fig. 5. OA on different numbers of PCA for four datasets.

A. Experimental Setup

For this experiment, we utilized the Adam optimizer and selected categorical cross-entropy as the loss function to train the suggested model. Assign a learning rate of 0.001 and weight decay of 0.0001. To evaluate the proposed model performance, we utilized the widely used HSI dataset Xuzhou, Pavia University, Salinas, and WHU-Hi-HC datasets for the training and testing, as employed with 0.5% training samples for Xuzhou, Pavia University and Salinas dataset and 0.05% training samples for WHU-Hi-HC datasets the validation set also used equal in size to the training set the remaining data was used for testing the model. The batch size and epoch for experiments are set to 64 and 100, respectively. We used principle component analysis (PCA) [51], [52] for the dimensionality reduction method, and we selected 25 spectral bands after applying PCA. The experiments were conducted ten times, and the mean and standard deviation values were evaluated and reported in the tables.

B. Model Parameter Selection

This section examines the factors that influence classification accuracy, such as the varying patch size or input window size.

1) Impact of Patch Size: In this experimental study, cubes of varying sizes are employed in the area of the central pixel input to maximize the utilization of spatial–spectral information in the HSI. Classification accuracy can be influenced by the input cubes. Therefore, we conducted experiments using various patch sizes and defined the input patch size for four datasets. The overall accuracy (OA) of the proposed model increases consistently as the patch size increases for three datasets. The primary reason is the variation of the pixel distribution among the three datasets. For example, the WHU-Hi-HC dataset shows scattered land cover areas, resulting in a decrease in OA for patch sizes larger than 13 × 13.

The smaller patch size captures local details well but may miss broader spatial context, leading to lower classification accuracy because the model does not have enough information about the surrounding area. The large patch size includes more surrounding pixels, which helps the model understand the context better, increasing accuracy. Fig. 4 shows the different input patch sizes on three datasets.

2) Number of PCA Components: Before employing the feature extractor module, PCA is applied to reduce the dimensionality of the data, thereby decreasing the number of model parameters. Experimental results demonstrate that the number of principal components in PCA significantly influences the extraction of spectral–spatial features. Additionally, we are conducting experiments to explore the correlation between the number of spectral bands and OA classification accuracy. The findings from these experiments are presented in Fig. 5.

3) Number of Transformer Encoders: To investigate the influence of the transformer encoders on classification performance, we conducted experiments by varying the number of

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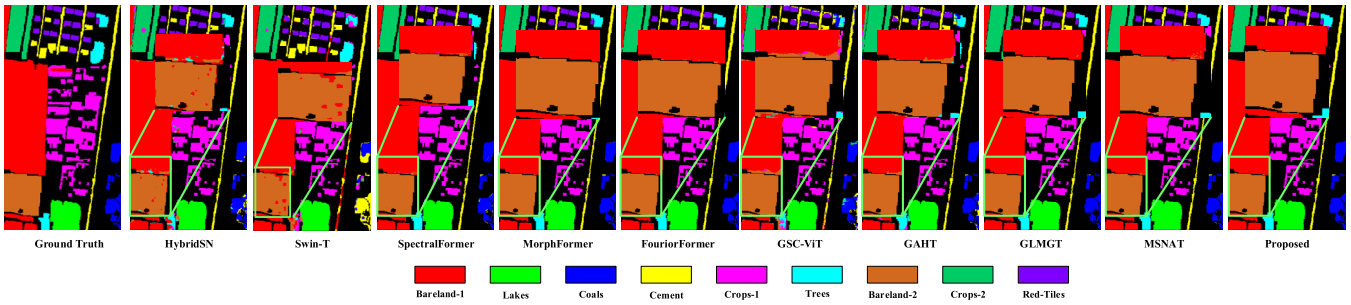


Fig. 6. Classification maps obtained by different models on the Xuzhou Dataset.

TABLE III
OA (%) OF PROPOSED MODEL WITH DIFFERENT NUMBERS OF TRANSFORMER ENCODERS ON FOUR DATASETS

Number of transformer encoder	XU	PU	SA	WHU-Hi-HC
1	98.58	97.79	98.45	96.43
2	**99.50**	**98.06**	**99.20**	**97.24**
3	96.97	97.45	97.58	97.18
4	97.24	96.98	98.10	96.32

encoders from 1 to 4. The results on four datasets are shown in Table III. It is evident that the optimal number of encoders is 2.

V. RESULT AND DISCUSSION

A. Classification Performance

To verify the effectiveness of the proposed model, experimental results of the state-of-the-art models are presented: HybridSN [17], Swin-T [42], SpectralFormer [29], MorphFormer [43], FouriorFormer [44], GSC-ViT [45], GLMGT [53], GAHT [46], and MSNAT [47]. For a fair comparison, we use the same training samples and the same patch size for the comparison models. We conducted the experiments on three benchmark datasets, the details of which are listed in Tables I and II. For the Xuzhou dataset, the experimental results of each model, including OA, average accuracy (AA), and Kappa and class-wise accuracy, are shown in Table IV. Fig. 6 represents the classification maps on the Xuzhou dataset of each model. HybridSN, Swin-T, and GSC-ViT show weak performance as they fail to effectively learn Features under small training data, leading to clear errors in their classification maps. HybridSN and GAHT achieve high OA but exhibit low AA due to poor performance in specific classes, such as Class 8 for GAHT, which results in blocky errors. HybridSN shows weak performance in Classes 3 and 6, revealing an unclear classification structure compared to the ground truth. Swin-T achieves below 90% accuracy in classes 3, 4, 6, and 9. In general, SpectralFormer, MorphFormer, FouriorFormer, GLMGT, and MSNAT perform well, reducing noisy pixels in the maps; however, GSC-ViT struggles with Classes 3, 4, and Class 6, achieving accuracy below 90%. Our proposed model approach surpasses other advanced techniques, achieving the highest AA, OA, and KAPPA scores, with all class accuracies exceeding 98%. Additionally, the classification map generated by our method is the most precise and smooth, closely resembling the ground-truth image. Table V shows the classification results for the Salinas dataset. The results revealed that the proposed model exhibits constant performance gain in all the classes. It can be observed that the OA achieved by the SpectralFormer and MorphFormer models is slightly low. Focusing on the GSC-ViT and GAHT models outperformed and gained the highest OA. The proposed model obtained better classification results in "Lettuce-romaine-7kw" and "Vineyard_untrained" where the cross-attention layer helps to capture features better. Compared to the second-best model, the proposed model achieved 0.10%, 0.28%, and 0.05% in terms of OA, AA, and the kappa coefficient. Fig. 7 shows the classification maps of different methods on the Salinas Dataset; as we can see, the GAHT model and proposed model have less noise. Our classification is almost near the ground-truth image. As the level of noise increases, the accuracy of the classification maps tends to decrease.

Table VII shows the result of all metrics on the WHI-Hi-HC dataset, and Fig. 9 shows the classification maps obtained by all models on the WHU-Hi-HC dataset. Our model demonstrates outstanding classification performance, achieving AA, OA, and KAPPA scores exceeding 97%, 94%, and 0.96, respectively—the highest among all compared methods. Specifically, the HybridSN, Swin-T, and SpectralFormer models continue to underperform due to using only 0.05% of labeled samples for training, and MSNAT does not classify as well as it did on the Xuzhou dataset. The basic reason for this is that the spectralFormer model does not fully exploit the 3-D nature of the data, and it is challenging to implicitly represent the spatial relationship with a limited number of training samples. Out of many comparative algorithms, FouriorFormer, GSC-ViT, GAHT, and GLMGT are particularly well suited for HSI data and show superior performance since they employ a convolutional layer. Hybrid CNN and transformer approaches outperform transformer approaches. This implies that the combination of two architectures can be advantageous, but it requires more refinement. Furthermore, the proposed model effectively extracts spectral spatial features specifically designed for HSI data. It successfully extracts both local and high-level features by combining the strength of the CNN and transformer-based attention module. As a result, it outperforms other transformer-based classification

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TABLE IV
CLASSIFICATION RESULT (%) ON XUZHOU DATASET

Class	HybridSN	Swin-T	SF	MorphFormer	FouriorFormer	GSC-ViT	GAHT	GLMGT	MSNAT	Proposed
C1	96.96(±1.04)	96.90(±0.14)	98.97(±0.03)	99.30(±0.07)	99.10(±0.09)	94.14(±0.32)	95.20(±0.8)	99.40(±0.06)	99.30(±0.7)	**99.62(±0.18)**
C2	96.91(±3.10)	94.40(±1.10)	99.40(±0.6)	99.80(±0.02)	**100**(±0.00)	99.22(±0.16)	98.05(±0.95)	99.97(±0.03)	99.38(±0.62)	99.80**(±0.02)**
C3	89.07(±2.93)	82.52(±0.04)	96.70(±0.3)	98.20(±1.08)	99.50(±0.05)	85.81(±1.26)	95.70(±1.03)	98.90(±1.01)	**99.68**(±0.32)	99.60**(±0.04)**
C4	96.45(±2.55)	75.39(±0.36)	99.30(±0.7)	99.70(±0.03)	99.58(±0.22)	89.28(±1.02)	97.30(±1.07)	99.50(±0.05)	99.79(±0.01)	**99.90(±0.01)**
C5	97.49(±1.51)	94.40(±2.04)	99.52(±0.48)	99.73(±0.17)	99.47(±0.13)	98.79(±0.02)	99.50(±0.05)	**99.70**(±0.03)	99.86(±0.11)	99.40**(±0.06)**
C6	88.64(±1.36)	84.78(±1.34)	99.50(±0.5)	98.90(±1.0)	**100**(±0.00)	82.68(±0.35)	91.60(±2.04)	99.63(±0.37)	99.77(±0.02)	99.95**(±0.05)**
C7	96.17(±0.83)	92.57(±1.09)	99.60(±0.4)	**100**(±0.00)	99.60(±0.04)	99.98(±0.02)	99.70(±0.03)	99.50(±0.05)	99.84(±0.16)	99.96**(±0.04)**
C8	98.24(±1.76)	93.70(±0.98)	99.76(±0.3)	99.80(±0.02)	99.14(±0.16)	96.15(±1.95)	**100**(±0.00)	99.90(±0.01)	99.19(±0.01)	**100(±0.00)**
C9	98.97(±1.03)	86.29(±0.34)	**99.75**(±0.25)	99.20(±0.08)	99.02(±0.18)	95.88(±0.37)	93.17(±1.83)	99.80(±0.20)	99.63(±0.07)	98.66**(±1.04)**
OA	96.53(±1.47)	92.50(±0.44)	99.18(±0.82)	99.47(±0.23)	99.45(±0.25)	95.41(±2.59)	96.98(±0.02)	99.45(±0.13)	99.55(±0.15)	**99.62(±0.18)**
AA	95.43(±1.57)	88.99(±0.67)	99.16(±0.84)	99.40(±0.30)	99.49(±0.11)	93.54(±3.46)	96.69(±0.31)	99.58(±0.02)	99.60(±0.40)	**99.65(±0.25)**
K×100	95.60(±1.40)	88.90(±3.78)	99.05(±0.95)	99.38(±0.42)	99.44(±0.16)	94.60(±2.40)	96.15(±1.85)	99.39(±0.11)	99.45(±0.35)	**99.49(±0.31)**
Param(K)	687.24K	893.87K	128.09K	141.9K	621.91K	**81.4K**	1.06M	104.67K	122.19K	485.12K
MACs(M)	0.0469M	0.146M	3.09M	7.19M	94.33M	8.9M	177.62M	12.38M	7.6M	**0.209M**
TR Time(s)	**323.67**	**401.12**	**392.45**	1854.18	2795.35	2019.51	2101.92	152.46	1613.13	**670.46**
TS Time(s)	23.05	27.69	31.92	32.72	18.56	24.06	26.41	19.93	**16.08**	113.65

TABLE V
CLASSIFICATION RESULT (%) ON SALINAS DATASET

Class	HybridSN	Swin-T	SF	MorphFormer	FouriorFormer	GSC-ViT	GAHT	GLMGT	MSNAT	Proposed
C1	99.59(±0.01)	53.69(±0.10)	97.94(±1.06)	99.49(±0.51)	100(±0.00)	100(±0.00)	99.49(±0.41)	100(±0.00)	99.65(±0.05)	100(±0.00)
C2	100(±0.00)	98.30(±0.78)	100(±0.00)	100(±0.00)	100(±0.00)	100(±0.00)	100(±0.00)	100(±0.00)	100(±0.00)	100(±0.00)
C3	99.84(±0.06)	84.20(±1.56)	99.94(±0.03)	100(±0.00)	99.51(±0.10)	100(±0.00)	100(±0.00)	100(±0.00)	100(±0.00)	99.94(±0.06)
C4	99.42(±0.08)	99.50(±0.30)	97.60(±1.40)	95.86(±1.14)	99.32(±0.05)	99.85(±0.15)	100(±0.00)	99.56(±0.44)	98.36(±1.04)	100(±0.00)
C5	98.79(±1.01)	98.57(±1.15)	97.54(±1.46)	93.43(±3.57)	99.88(±0.12)	96.79(±0.21)	98.83(±1.07)	96.71(±1.29)	94.54(±2.46)	99.47(±0.53)
C6	100(±0.00)	100(±0.00)	100(±0.00)	99.97(±0.03)	100(±0.00)	100(±0.00)	100(±0.00)	100(±0.00)	99.33(±0.67)	99.89(±0.01)
C7	99.94(±0.02)	97.69(±2.36)	99.54(±0.36)	99.46(±0.54)	100(±0.00)	99.97(±0.03)	99.97(±0.03)	100(±0.00)	100(±0.00)	100(±0.00)
C8	95.94(±1.06)	90.72(±1.26)	94.55(±1.45)	97.96(±1.04)	99.73(±0.27)	98.67(±0.33)	99.36(±0.04)	95.16(±1.84)	94.08(±2.92)	98.18(±0.82)
C9	100(±0.00)	99.96(±0.04)	100(±0.00)	99.59(±0.41)	100(±0.00)	99.98(±0.02)	100(±0.00)	100(±0.00)	100(±0.00)	100(±0.00)
C10	96.64(±1.36)	97.01(±1.16)	96.73(±0.27)	99.35(±0.65)	99.44(±0.56)	99.84(±0.06)	99.72(±0.08)	99.01(±0.99)	98.17(±0.83)	99.81(±0.09)
C11	99.05(±0.35)	84.68(±0.46)	97.72(±1.28)	99.24(±0.76)	99.81(±0.19)	99.33(±0.37)	98.31(±1.09)	99.90(±0.10)	98.90(±0.10)	**100**(±0.00)
C12	99.52(±0.28)	97.79(±0.56)	98.89(±0.11)	99.73(±0.27)	100(±0.00)	100(±0.00)	99.58(±0.02)	100(±0.00)	100(±0.00)	100(±0.00)
C13	97.02(±1.98)	96.87(±1.36)	94.27(±1.73)	93.18(±2.82)	80.51(±7.49)	98.89(±0.11)	94.15(±1.85)	90.45(±1.55)	94.45(±0.55)	97.60(±1.00)
C14	96.97(±1.03)	97.98(±2.06)	97.16(±0.84)	98.20(±0.80)	97.92(±0.08)	98.11(±0.89)	91.78(±1.22)	99.33(±0.67)	98.54(±0.46)	**99.81**(±0.10)
C15	90.21(±5.79)	86.93(±1.54)	92.87(±5.13)	98.19(±0.81)	99.01(±0.09)	98.13(±0.87)	98.99(±1.01)	99.19(±0.81)	98.21(±0.79)	**99.30**(±1.81)
C16	99.32(±0.38)	94.58(±1.30)	99.04(±0.56)	99.77(±0.23)	99.66(±0.34)	99.83(±0.17)	100(±0.00)	99.27(±0.73)	99.20(±0.80)	100(±0.00)
OA	97.40(±1.60)	93.27(±0.66)	96.94(±0.06)	98.50(±0.50)	98.97(±0.50)	99.20(±0.40)	99.15(±0.75)	98.34(±0.66)	98.38(±0.62)	**99.30**(±0.60)
AA	98.27(±0.73)	92.40 (±2.78)	97.74(±0.26)	98.34(±0.66)	98.42(±1.00)	99.34(±0.36)	98.76(±0.24)	98.66(±0.34)	98.23(±0.77)	**99.62**(±0.30)
K×100	97.10(±1.90)	92.27(±2.80)	96.66(±0.34)	98.30(±0.06)	98.88(±0.35)	99.15(±0.25)	99.12(±0.78)	98.15(±0.85)	98.19(±0.81)	**99.20**(±0.50)
Param(K)	534.66K	658.78K	128.54K	142.45K	621.74K	**81.97K**	1.06M	115.5K	122.06K	587.79K
MACs(M)	16.11M	16.89M	3.09M	7.19M	94.33M	12.58M	177.62M	12.39M	7.7M	**0.209M**
TR Time(s)	**162.75**	**251.25**	**208.41**	699.58	645.54	601.43	820.09	842.11	629.11	**459.87**
TS Time(s)	37.70	45.87	78.09	160.08	79.89	85.51	86.61	124.10	**40.71**	101.24

models, providing a significant advantage. Comparison models may exhibit poor performance when HSI classification is used with a limited number of samples. A good example is the SpectralFormer shows a lack of accuracy, scoring 41% in category 6 on the WHU-Hi-HC dataset.

The classification results of Pavia University are shown in Table VI. Hybrid CNN and transformer approaches outperform transformer approaches. This implies that the combination of two architectures can be advantageous, but it requires more refinement. It successfully extracts both local and high-level features by combining the strength of the CNN and transformer-based attention module. As a result, it outperforms other transformer-based classification models, providing a significant advantage. Comparison models may exhibit poor

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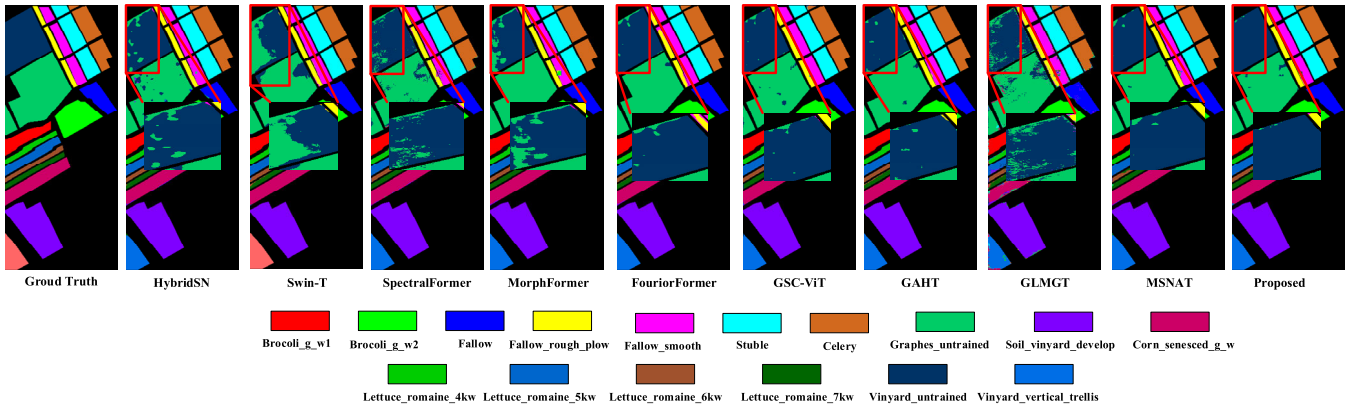


Fig. 7. Classification maps obtained by different models on the Salinas Dataset.

TABLE VI
CLASSIFICATION RESULT (%) ON PAVIA UNIVERSITY DATASET

Class	HybridSN	Swin-T	SF	MorphFormer	FourierFormer	GSC-ViT	GAHT	GLMGT	MSNAT	Proposed
C1	94.30(±1.70)	94.71(±0.60)	93.75(±2.25)	90.92(±1.08)	99.04(±0.56)	94.69(±1.31)	96.00(±1.00)	99.42(±0.01)	99.21(±0.39)	96.47(±1.53)
C2	99.61(±0.04)	99.52(±0.35)	99.94(±0.06)	99.74(±0.26)	100(±0.00)	99.71(±0.09)	99.58(±0.42)	100(±0.00)	100(±0.00)	99.89(±0.11)
C3	80.98(±1.02)	89.97(±0.72)	70.33(±2.50)	90.16(±1.84)	90.54(±0.46)	90.39(±1.61)	91.89(±6.11)	88.80(±1.02)	91.80(±6.10)	**95.20**(±2.80)
C4	82.75(±1.25)	91.79(±2.33)	89.31(±1.60)	80.01(±1.89)	82.22(±0.77)	92.47(±2.53)	90.26(±2.74)	94.88(±5.10)	90.88(±3.01)	91.70(±5.30)
C5	100(±0.00)	98.63(±1.28)	96.39(±1.61)	98.79(±1.01)	100(±0.00)	99.47(±0.03)	99.54(±0.46)	100(±0.00)	100(±0.00)	100(±0.00)
C6	97.53(±1.47)	98.88(±0.56)	88.82(±1.10)	99.01(±0.89)	100(±0.00)	99.65(±0.05)	97.97(±2.03)	99.21(±0.63)	98.21(±1.19)	99.95(±0.05)
C7	82.83(±1.20)	98.89(±0.65)	70.34(±2.06)	96.43(±1.07)	97.57(±0.43)	98.86(±1.04)	98.57(±1.03)	85.64(±1.30)	92.64(±5.36)	**100**(±0.00)
C8	84.40(±1.60)	92.34(±2.70)	80.62(±1.30)	85.43(±1.05)	95.06(±1.94)	92.46(±5.54)	97.25(±2.05)	90.97(±8.03)	81.97(±10.03)	**97.80**(±1.20)
C9	96.70(±0.30)	89.99(±0.78)	81.00(±1.02)	84.67(±1.03)	82.08(±1.09)	90.60(±3.40)	91.65(±5.35)	89.26(±0.7)	90.25(±4.75)	94.15(±5.85)
OA	94.10(±1.90)	96.25(±0.64)	88.42(±8.05)	93.40(±1.60)	96.60(±1.40)	96.40(±1.60)	96.45(±1.55)	94.35(±2.65)	94.10(±3.90)	**98.19**(±0.81)
AA	91.01(±5.99)	94.96 (±0.12)	85.61(±1.03)	91.68(±2.32)	94.06(±3.94)	95.37(±2.63)	95.86(±2.14)	94.24(±4.76)	93.88(±4.12)	**97.24**(±0.76)
K×100	92.09(±3.91)	95.13(±0.58)	84.27(±1.07)	91.19(±3.81)	95.48(±2.52)	95.20(±2.80)	94.52(±2.48)	93.10(±5.90)	92.08(±6.92)	**97.65**(±0.35)
Param(K)	687.24K	839.68K	128.09K	141.9K	621.91K	**81.4K**	1.06M	104.67K	122.19K	485.12K
MACs(M)	0.0469M	0.131M	3.09M	7.19M	94.33M	8.9M	177.62M	12.38M	7.6M	**0.209M**
TR Time(s)	**310.51**	**420.69**	**388.35**	1798.20	1884.40	2120.62	2098.88	198.21	1512.10	390.36
TS Time(s)	19.15	25.34	32.81	35.61	19.67	30.10	29.50	19.82	**20.36**	57.29

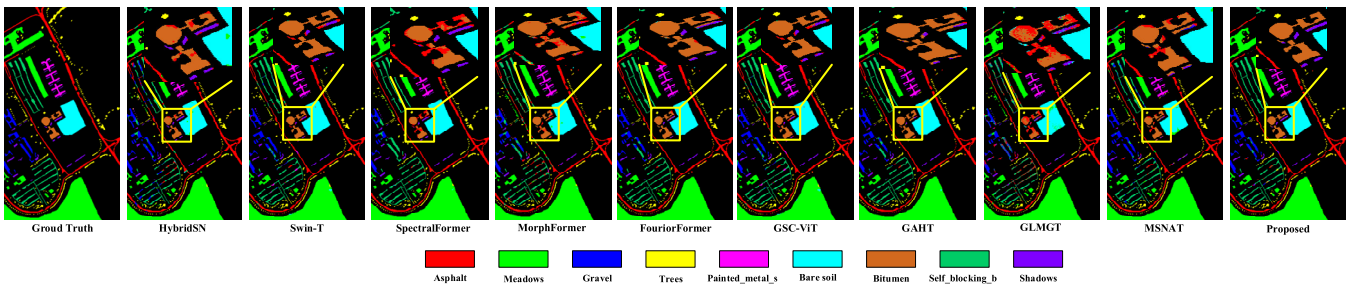


Fig. 8. Classification maps obtained by different models on the Pavia University dataset.

performance while HSI classification uses a limited number of samples. The proposed model achieves the best classification result in all classes, specifically in class "Gravel," "Bitumen," and "self-blocking" achieved best classification results. Fig. 8 shows the classification maps on the Pavia University dataset. As one can see, GSC-ViT, GAHT, and FourierFormer obtain excellent classification results with less noise and intraclass smoothness.

B. Model Complexity Analysis

Tables IV–VII show the model complexity, revealing that the proposed model is less complex in terms of FLOPs. The parameters (MACs) of the proposed model are higher than GAHT and GLMGT models. The parameter of the model is slightly higher because to attention layer value is 2. As we can see, the training time of the proposed model is lower than other transformer-based models, except that SpectralFormer didn't

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TABLE VII
CLASSIFICATION RESULT (%) ON WHU-HI-HANCHUAN DATASET

Class	HybridSN	Swin-T	SF	MorphFormer	FouriorFormer	GSC-ViT	GAHT	GLMGT	MSNAT	Proposed
C1	96.99(±1.01)	98.50(±0.25)	95.46(±4.44)	99.61(±0.39)	99.51(±0.49)	98.94(±1.06)	99.16(±0.84)	98.78(±1.22)	98.04(±0.96)	**99.39**(±0.01)
C2	94.38(±2.62)	92.25(±3.61)	89.58(±1.04)	93.78(±6.22)	93.30(±5.70)	94.06(±5.94)	94.13(±0.87)	**95.04**(±4.96)	92.08(±0.82)	94.57(±2.43)
C3	90.96(±5.04)	96.70(±2.11)	56.84(±3.31)	88.09(±11.9)	97.74(±2.26)	96.67(±3.33)	93.13(±1.87)	96.14(±3.86)	88.53(±1.14)	**99.52**(±0.08)
C4	96.35(±2.65)	98.57(±1.10)	96.84(±3.16)	98.28(±1.72)	98.92(±1.08)	98.67(±1.33)	**99.18**(±0.82)	98.82(±1.18)	97.20(±2.80)	98.62(±1.08)
C5	81.01(±1.09)	86.97(±0.15)	80.24(±1.97)	83.08(±1.59)	87.03(±1.29)	88.47(±1.15)	**94.10**(±2.90)	74.64(±2.43)	82.54(±1.74)	89.56(±1.04)
C6	85.86(±1.01)	81.01(±0.64)	41.62(±5.83)	69.20(±3.08)	80.04(±1.99)	86.29(±1.37)	83.73(±1.62)	72.27(±2.67)	75.13(±2.48)	**86.94**(±1.30)
C7	79.65(±5.30)	93.87(±2.31)	47.48(±5.25)	91.35(±8.65)	94.50(±5.50)	92.69(±5.31)	91.51(±3.49)	89.08(±1.09)	90.46(±4.54)	**96.14**(±0.56)
C8	89.47(±7.50)	95.98(±0.06)	84.37(±5.56)	96.53(±3.47)	**96.94**(±3.06)	95.76(±3.24)	96.61(±3.39)	90.72(±2.98)	88.87(±1.01)	94.61(±0.39)
C9	80.56(±1.40)	94.67(±3.17)	69.88(±9.91)	80.79(±9.20)	95.93(±4.07)	97.65(±0.35)	91.70(±2.30)	93.16(±0.84)	85.93(±1.40)	**98.42**(±0.58)
C10	97.06(±0.94)	96.88(±0.64)	85.97(±1.40)	98.95(±1.05)	97.78(±2.22)	98.87(±0.13)	**99.15**(±0.85)	97.99(±0.01)	95.89(±4.11)	98.53(±0.47)
C11	93.54(±0.46)	96.79(±1.36)	87.48(±6.25)	97.78(±2.22)	97.98(±2.02)	97.46(±0.54)	95.77(±0.23)	94.58(±0.42)	96.81(±3.19)	**98.67**(±0.33)
C12	57.27(±9.70)	87.90(±0.20)	71.80(±8.82)	73.14(±2.68)	88.17(±1.18)	84.84(±1.51)	77.81(±2.01)	65.85(±4.31)	79.54(±2.04)	**93.95**(±1.05)
C13	70.50(±9.50)	84.12(±0.39)	54.81(±8.51)	72.12(±2.78)	85.83(±1.41)	**88.27**(±1.17)	80.45(±1.95)	80.17(±1.88)	71.39(±2.06)	83.51(±0.44)
C14	92.18(±3.82)	95.69(±0.88)	85.47(±9.45)	94.01(±5.99)	96.09(±3.91)	95.28(±0.72)	95.89(±0.11)	94.29(±0.71)	92.06(±0.94)	**97.30**(±2.10)
C15	56.08(±4.90)	82.87(±1.47)	42.40(±7.56)	61.95(±3.80)	83.42(±1.65)	83.26(±1.67)	74.08(±2.59)	76.84(±2.31)	81.57(±1.74)	**85.24**(±5.07)
C16	99.66(±0.04)	98.99(±1.10)	98.62(±1.38)	99.69(±0.31)	99.89(±0.11)	99.86(±0.14)	**99.91**(±0.09)	99.67(±0.33)	99.61(±0.39)	99.87(±0.03)
OA	92.50(±6.50)	96.69(±1.05)	85.70(±9.43)	94.88(±5.12)	96.85(±3.15)	96.90(±0.10)	95.78(±0.22)	94.94(±1.06)	93.10(±4.00)	**97.39**(±2.21)
AA	85.09(±4.09)	92.61(±1.25)	74.30(±5.27)	87.40(±1.06)	93.32(±5.68)	93.56(±2.44)	91.64(±3.36)	88.63(±1.03)	88.48(±1.05)	**94.68**(±0.32)
K×100	91.19(±5.11)	95.69(±0.57)	83.20(±6.80)	93.98(±5.02)	96.33(±1.67)	96.40(±1.60)	95.09(±0.91)	94.10(±0.90)	91.90(±1.10)	**96.98**(±0.02)
Param(K)	534.66K	682.36K	128.54K	142.45K	621.74K	**81.97K**	1.06M	115.5K	122.06K	587.79K
MACs(M)	16.11M	16.92M	3.09M	7.19M	94.33M	12.58M	177.62M	12.39M	7.7M	**0.209M**
TR Time(s)	**151.87**	**213.78**	**210.33**	706.13	658.67	544.16	807.15	839.55	636.12	**506.69**
TS Time(s)	39.76	42.55	77.17	153.18	82.59	84.66	85.73	121.01	**39.70**	110.51

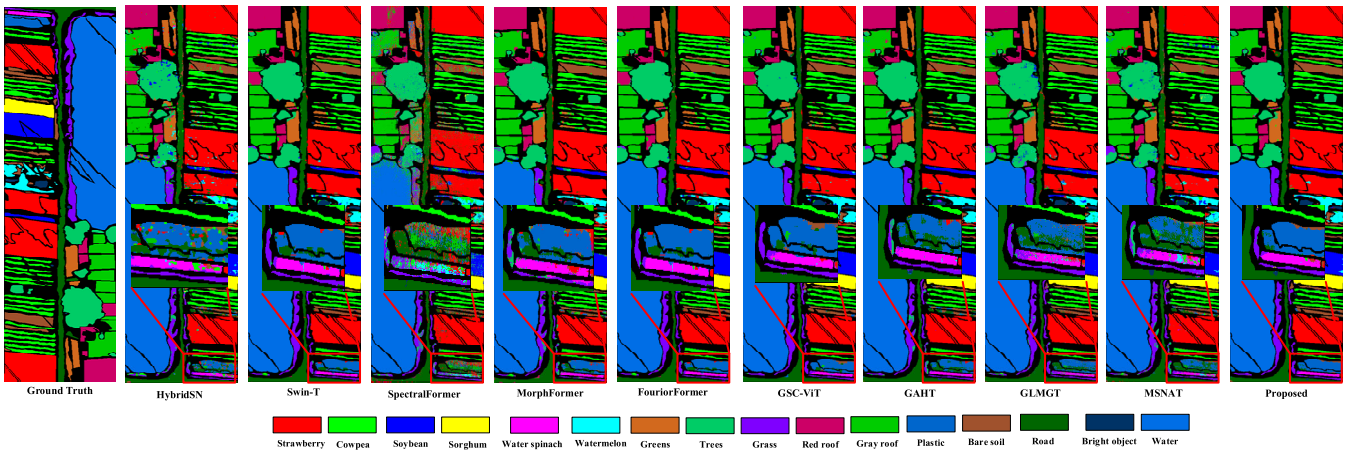


Fig. 9. Classification maps obtained by different models on the WHU-Hi-HC dataset.

use any convolutional layer. However, in terms of testing time MSNAT model performs well the parameters of this model are also very low.

We also report the training and testing time obtained by different models on four datasets. As shown in Tables IV–VII, the proposed model has the highest parameters among other models, but also achieves significantly shorter training and testing time than some other models, MorphFormer, FouriorFormer, GSC_ViT, and MSNAT models. SpectralFormer and HybridSN models demonstrated that they have faster speed while maintaining performance ≥85%. On the Pavia University dataset, the proposed model completes training in 390.36 s, which is faster than GSC_ViT and GAHT models. The t-SNE dimensionality reduction technique enables the visualization of high-dimensional data in a lower-dimensional space. As illustrated in Fig. 10, samples belonging to the same class form distinct clusters, while samples from different classes are clearly separated from one another.

C. Ablation Experiments

To comprehensively highlight the efficacy of each module, we conducted various combinations on the four datasets using a patch size of 13 × 13, 0.5% and 0.05% training samples.

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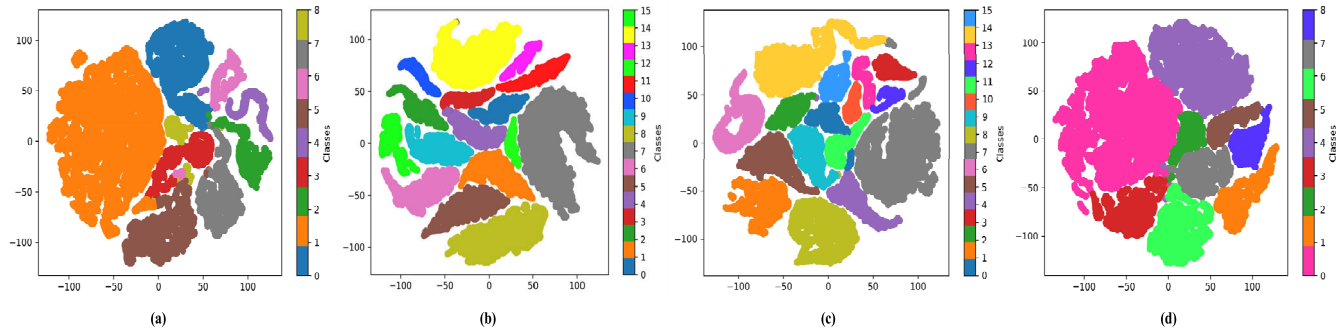


Fig. 10. t-SNE visualization on (a) Pavia University, (b) Salinas, (c) WHU-Hi-HC, and (d) Xzhou Dataset.

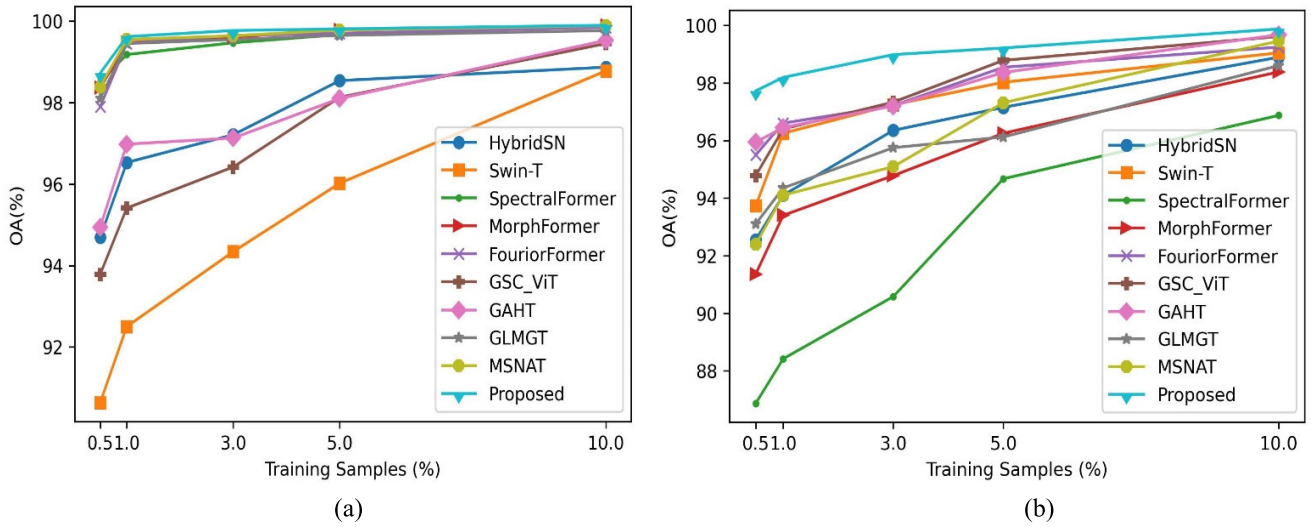


Fig. 11. Classification performance across different training samples. (a) Xuzhou dataset. (b) Pavia University Dataset.

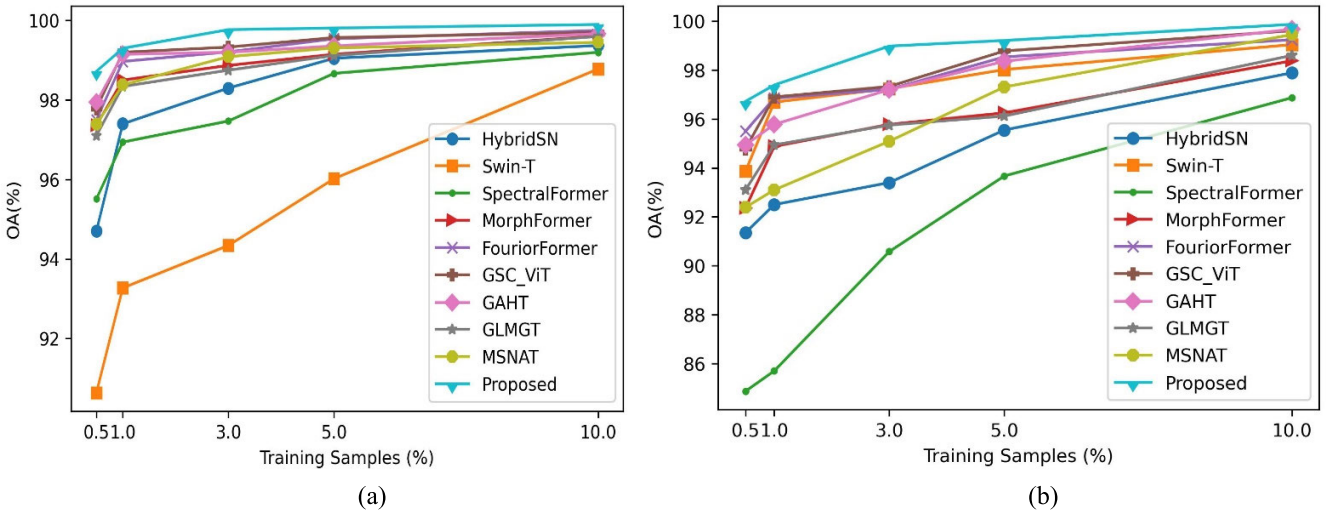


Fig. 12. Classification performance across different training samples. (a) Salinas dataset. (b) WHU-Hi-HC Dataset.

Our analysis focuses on the convolutional neural network, transformer module, and the integration of CNN, attention module, and IFE module, accessed through OA metrics. The outcomes of these experiments are presented in Table VIII. Overall, the proposed network can achieve the best performance results. Therefore, based on our network, it can be seen that when integrating cross cross-window attention module OA, AA, and Kappa increased by 0.38%, 1.44% and 3.13% on the Xuzhou dataset is useful to extract intricate features. These improvements highlight the effectiveness of the cross-window attention and IFE module in enhancing the model's capability to capture spatial complexities of hyperspectral data, resulting

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TABLE VIII
ABLATION EXPERIMENTS (%) WITH DIFFERENT MODULES ON FOUR DATASETS

Dataset	Components			Indicator
	Fused-MBConv	Cross Window	IFE	OA (%)
Xuzhou	✓	×	×	96.87
	✓	✓	×	98.56
	✓	✓	✓	**99.62**
	×	✓	✓	97.12
Pavia	✓	×	×	96.51
	✓	✓	×	97.87
	✓	✓	✓	**98.19**
	×	✓	✓	95.34
Salinas	✓	×	×	98.12
	✓	✓	×	98.93
	✓	✓	✓	**99.30**
	×	✓	✓	97.26
WHU-Hi-HC	✓	×	×	95.24
	✓	✓	×	96.37
	✓	✓	✓	**97.39**
	×	✓	✓	94.14

in better classification performance. In conclusion, the ablation experiments clearly demonstrate the substantial impact of each module in improving the model performance, especially in addressing spatial complexities and enhancing overall classification accuracy.

D. Impact of Training Samples

In order to evaluate the stability and robustness of the proposed model, we employ a random selection of training samples at various intervals. Figs. 11 and 12 show the OA of all comparison models with the proposed model on different training samples. As we can see, when the training samples increase, OA increases. Furthermore, it can be observed that the proposed model stabilized faster than other models. GSC_ViT, MorphFormer, GAHT, and FouriorFormer also stabilize, but at different OA accuracy points. Notably, our proposed model consistently achieves the highest accuracy, particularly when the training ratio is low, demonstrating clear superiority.

VI. CONCLUSION

In this work, we propose a novel HSI classification model incorporating a fused-MBConv module, cross-window attention block, and IFE to effectively extract deep spectral–spatial features. The fused-MBConv module serves as a robust feature extractor, leveraging an inverted residual structure to capture essential low-level spectral–spatial representations. To enhance long-range dependencies, the cross-window attention module enables feature aggregation across nonadjacent regions, addressing the limitations of conventional self-attention mechanisms constrained to local windows. Furthermore, the IFE module refines extracted features by facilitating interaction between spectral and spatial information, leading to more discriminative representations. Experimental evaluations on benchmark HSI datasets validate the superiority of the proposed model, demonstrating enhanced classification accuracy and robustness over state-of-the-art methods. To further improve the capability of long-range dependencies, our future plan is to investigate a dynamic or shifted grouping mechanism, inspired by the shifted window approach in Swin Transformer.

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